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The landscape genetics of syntopic topminnows (*Fundulus notatus* and *F. olivaceus*) in a riverine contact zone

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Abstract – The metapopulation dynamics of a species are influenced by the spatial arrangement of preferred habitats across a landscape. Increasing habitat patchiness can result in greater population subdivision, leading to more pronounced population genetic structure. Therefore, even subtle differences in habitat preferences among species could lead to substantial differences in genetic structure. Members of the *Fundulus notatus* species complex of topminnows include a headwater specialist and a large river specialist with broadly overlapping geographical distributions. Both species are often found in the same drainage systems, but partially isolated by divergent habitat preferences. The purpose of this research was to determine whether distribution along a stream continuum in a contact zone influences population genetic patterns in two species of topminnow (*Fundulus olivaceus* and *Fundulus notatus*). Fish were collected in the summers of 2010–2012, and habitat data were collected in summer 2012, within all major tributaries of the Saline River in southern Illinois. In total, 523 fish were genotyped using seven microsatellite loci to assess genetic variation. Species distributions in *F. notatus* and *F. olivaceus* were associated with habitat factors that covaried with cumulative drainage area. Populations of *F. olivaceus* showed a greater level of genetic structure reflected in a significant pattern of isolation by distance, while populations of *F. notatus* showed no such pattern. Spatial autocorrelation analysis revealed that at an individual level, both species became increasingly divergent with waterway distance. Overall, clear interspecific differences in habitat preferences were translated into measurable but relatively subtle differences in population genetic structure.

Key words: killifish; landscape genetics; isolation by distance; Fundulus; contact zone

Introduction

Genetic structure within populations can arise due to a myriad of factors, including organismal biology, habitat availability and suitability, and stochastic environmental processes. Landscape features can create habitat barriers, which disrupt dispersal of individuals between populations and generate genetic structure. Landscape features affect dispersal capabilities in terrestrial taxa including insects (Keyghobadi et al. 1999; Kanuch et al. 2012), birds (Piertney et al. 2008) and even humans (Sokal & Thomson 1998). In riverine systems, physical barriers such as waterfalls (Gomez-Uchida et al. 2009), artificial obstructions such as dams or weirs (Horreo et al. 2011), and poor habitat patches (Stelkens et al. 2012) can serve as barriers to gene flow in fish. Even in the absence of physical barriers, there can be isolation by distance, where populations become less genetically similar with growing geographical distance. Waterway distance between populations is correlated with a decrease in contemporary migration in multiple species and in multiple drainages (in trout: Carlsson et al. 1999; Kanno et al. 2011; Stelkens et al. 2012; in a variety of salmonids: Gomez-Uchida et al. 2009). Migration distances can vary by a large degree

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among populations of migratory fish such as trout, and these dispersal distances can be affected by environmental factors (Heggenes et al. 2006; Neville et al. 2006; Northcote 2010; Olsen et al. 2010).

The *Fundulus notatus* species complex provides opportunities to evaluate the influence of landscape on the genetic structure of populations. The species complex consists of three closely related killifish species that share similar ecological roles and overlapping ranges (Blanchard 1996). The blackstripe topminnow (*Fundulus notatus*) and the blackspotted topminnow (*Fundulus olivaceus*) have large ranges stretching from the Gulf Coast streams from Texas to western Florida, northward to southern Illinois and Missouri for *F. olivaceus*, and southern Wisconsin and Michigan for *F. notatus* (Thomerson 1966). The third member of this species complex, the broadstripe topminnow (*Fundulus euryzonus*), is endemic to the Lake Ponchartrain drainage of Louisiana and Mississippi (Suttkus & Cashner 1981). In natural contact zones, hybridisation is usually rare (Thomerson 1967; Duvernell et al. 2007), but can vary substantially among drainages (Schaefer et al. 2011; Duvernell & Schaefer 2013).

Fundulus notatus and *F. olivaceus* are found in different habitats when examined on a whole-stream landscape level. *Fundulus notatus* are typically found in downstream habitats with relatively low stream flow, mud or clay substrate and with aquatic vegetation. *Fundulus olivaceus* are generally more abundant in typical headwater habitats with shallower and narrower channels, lower bank slope, larger substrate size, less emergent vegetation, more woody debris and increased canopy cover (Braasch & Smith 1965; Blanchard 1996; Smiley et al. 2005; Schaefer et al. 2009; Duvernell & Schaefer 2013). Previous investigations suggest that relative abundance of these species within contact zones is affected by ecological variation including local microhabitat factors such as substrate and amount of aquatic vegetation (Braasch & Smith 1965; Schaefer et al. 2011). Larger-scale landscape variables including position within the stream continuum may also affect relative abundance (Schaefer et al. 2009) and fish community structure (Hitt & Angermeier 2008), especially when taking strength of ecological gradients along the river continuum and contact zone structure into account (Schaefer et al. 2011). Braasch and Smith (1965) suggested that the differences in habitat preference are due to relatively stable habitat factors such as stream velocity, substrate and aquatic vegetation rather than factors that change regularly such as turbidity and dissolved oxygen.

Given that *F. notatus* and *F. olivaceus* exist in sympatry, we wanted to determine whether interspecific differences in habitat use or association translated into different drainage-wide genetic structure patterns. To do this, we sampled from representative tributaries in each of the three forks and the main stem of the Saline River, a tributary of the Ohio River in southeastern Illinois, USA. This allowed us to assess drainage-wide genetic structure in each species and examine relationships between genetic and geographical distance, species distributions and habitat associations. Stream systems are dendritic; therefore, upstream locations are more isolated (i.e., headwater segments are separated by larger river habitats) from one another than downstream locations. Because of this inherent difference between the habitats preferred by each species, we hypothesised that differences in species distributions would translate into a greater level of genetic structure in the headwater species, *F. olivaceus*, than the larger river species, *F. notatus.* To examine how species distributions related to the landscape of the Saline River watershed, we examined habitat parameters at points of fish capture. We hypothesised that species relative abundance would be associated with habitat factors that correlated with stream size and relative position in the network (cumulative drainage area).

Methods

Site selection and sample collection

The Saline River drainage in south-eastern Illinois, USA, marks the northern limit of *F. olivaceus* in the eastern portion of its range in the Ohio River Basin. The Saline River lies along the border between the Shawnee Hills upland region and the Mount Vernon Hill Country prairie region (Leighton et al. 1948). *Fundulus olivaceus* is abundant in the small upland drainages of the Shawnee Hill country (Braasch & Smith 1965; Duvernell et al. 2007), whereas *F. notatus* is abundant in prairie streams further to the north (Braasch & Smith 1965).

Fish were collected from 21 sites covering the three major tributary forks as well as along the main stem of the Saline River in the summers of 2010–2012. At each site, fish were sampled using dip net or seine. A caudal fin clip was taken from each fish for further DNA analysis, and fish were either preserved in formalin for morphological analysis in a separate study (Duvernell & Schaefer 2013) or returned to the stream at the point of capture.

Habitat assessment

Location of each sampling site was recorded using global positioning system (GPS) coordinates. Habitat parameters were measured at six points (three on the left bank and three on the right bank) surrounding a

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location within a site where fish were captured. The habitat parameters measured were water depth (in cm), stream width (cm), flow rate (m/s), substrate size (modified Wentworth scale after Cummins 1962), presence or absence of aquatic vegetation and in-stream cover, bank stability and slope, and canopy cover. Cumulative drainage area was extracted from the National Hydrology Plus database (http://www.horizon-systems.com/nhdplus) as a descriptor of stream size and local hydrology.

Habitat variables were averaged for each site and summarised with a principal component analysis (PCA) of the correlation matrix. Cumulative drainage area and the first three principal components were tested for multicollinearity using Pearson's r. Parameters that showed correlations greater than 0.2 were entered into a stepwise regression to determine relative importance in predicting proportion of species. All statistical evaluations of habitat parameters were performed using SYSTAT, version 10 (Systat software, Chicago, IL, USA).

Genetic structure

Individuals were genotyped using seven microsatellite loci – Fno112, FhATG-20, FhATG-6, FhATG-B103, Fno091, FhATG-B101 and Fno056 – following the studies by Duvernell et al. (2008) and Feldheim, K.A., Kreiser, B.R., Schmidt, B., Duvernell, D.D. & Schaefer, J.F. (in review) (Table 1). Polymerase chain reactions (10 μl volume) for microsatellite analysis contained approximately 50 ng DNA, 200 mM of each dNTP, 1 μM of each primer and 1 unit of Taq polymerase (Sigma-Aldrich Inc., St. Louis, MO, USA) in 10× buffer (1.5 mM of $MgCl_2$). Thermal cycling conditions included an initial denature step of 95 °C for 5 min followed by 10 cycles of 94 °C for 30 s, 56 °C for 45 s, 72 °C for 45 s, then 20 cycles of 94 °C for 30 s, 53 °C for 45 s, 72 °C for 45 s and a final step of 72 °C for 10 min. Primers were fluorescently labelled and amplified products were analysed using an Applied Biosystems 3730 × l (Applied Biosystems Inc., Foster City, CA, USA). Electropherograms were analysed using GENEMAPPER, version 4.0 (Applied Biosystems Inc.).

Table 1. Forward (F) and reverse (R) primers used to amplify microsatellite sequences in this study. Locus names with Fno were developed using *Fundulus notatus* genomic sequences (Feldheim, K.A., Kreiser, B.R., Schmidt, B., Duvernell, D.D. & Schaefer, J.F. in review). Fh loci were developed using *F. heteroclitus* genomic sequences (Duvernell et al. 2008). One primer was developed in *F. heteroclitus*, and then a modified reverse primer was optimised using *F. notatus* sequence.

Locus	Sequence (5′–3′)	GenBank accession number
Fno056F	GACCAACCGTTTGTCAATGCAAC	KF702366
Fno056R	AACCCACATGTTCAAGACTGC	KF702366
Fno091F	TCAATAGATAAGGTCATTTGTTTTCAA	KF702367
Fno091R	AAAGCTTATTTATCCTTCCACTGC	KF792367
Fno112F	TGGGTGTGAGTGCATATGGT	KF702370
Fno112R	CCGTTAAGGCCAGTGTTCAT	KF702370
FhATG-B103F	CGGAGCATTGTGATTGTGTTGTTTT	AY791487
FhATG (Fno)-B103R	AAGGTTGTAAAATGGGACAGG	AY791487
FhATG-B101F	ACCCGTATGGATTTAAAAGCAACC	AY791485
FhATG-B101R	GCCGCTCGACCGCCTGAAC	AY791485
FhATG-20F	GGTTTAACTTGGACCGAGTCTCC	AY791455
FhATG-20R	AGAGTAGTAACTGGGTCGCTGATTT	AY791455
FhATG-6F	AACATGAGAACCCCGCCCACC	AY791444
FhATG-6R	CACAAGTTCAACTGTATATCCTCTCC	AY791444

Raw microsatellite genotypes were converted to STRUCTURE format using CONVERT, version 1.31 (Glaubitz 2004). Individuals were assigned to species using STRUCTURE, version 2.3 (Pritchard et al. 2000). Modelling parameters used were $k = 2$, 1,00,000 MCMC repetitions and a burn in period of 10,000. Individuals were assigned as pure species (no hybrid ancestry) if they exhibited an admixture proportion greater than 0.95 to either *F. olivaceus* or *F. notatus*. Individuals with admixture values less than 0.95 were treated as putative hybrids and not used in subsequent analyses. Determination of genetic structure within species was performed using STRUCTURE, version 2.3. Modelling parameters used were $k = 1$ through $k = 20$ with 1,00,000 MCMC repetitions and a burnin of 10,000. This analysis was repeated 10 times per k value. The Evanno method was used to determine the most likely k for each species and was carried out using STRUCTURE HARVESTER, version A.1 (Evanno et al. 2005; Earl & vonHoldt 2012). Analysis of molecular variance (AMOVA) was performed for each species using ARLEQUIN, version 3.5 (Excoffier & Lischer 2010).

Genetic and geographical distance correlations

Determination of F_{ST} values for each pair of populations within species and tests for Hardy–Weinberg equilibrium were performed using GENEPOP, version 4.0 (Raymond & Rousset 1995; Rousset 2008). Geographical distance between each population-pair was determined by plotting GPS coordinates for each collection site and measuring river distance (km) in GOOGLE EARTH (Google, Mountain View, CA). Mantel tests were performed on genetic and geographical distances using the Isolation by Distance Web Service, version 3.23 (Jensen et al. 2005), with 10,000 randomisations. Pairwise habitat distances were compared by computing Euclidean distance between site PC scores. These scores were used in partial Mantel tests with genetic and geographical distance to control for the effects of habitat differ-

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ences using ZT, version 1.1 (Bonnet & Van de Peer 2002).

Comparisons of genetic and geographical distance at the level of the individual were made using the spatial autocorrelation function in GenAlEx, version 6.5 (Peakall & Smouse 2012). This test used the squared distances measure to estimate genetic distances between pairs of individuals and then compared those genetic distances to the same waterway distances used for Mantel tests. Individuals collected from the same site were assigned a waterway distance of zero. To assess nonrandom genetic structure at the individual level, a correlogram of the autocorrelation coefficient indicating relatedness of individuals (r) was plotted as a function of distance class using 15 km increments.

Results

We collected *Fundulus* sp. individuals and measured habitat variables at seventeen sites throughout the Saline River drainage. Sample sizes for each species at each site ranged from 6 to 93 with a median sample size of 25 (Table 2), and a total of 228 *F. notatus*, 258 *F. olivaceus* and 37 individuals of putative hybrid ancestry were identified.

Fundulus olivaceus individuals were most abundant in upstream reaches of each tributary fork with *F. notatus* individuals more common in the larger river habitats. However, both species were present at many sampled locations, and *F. olivaceus* individuals were absent in only two sites throughout the drainage (Fig. 1). *F. notatus* individuals were absent from five sampled sites, all occurring in the smallest headwater sites with the lowest cumulative drainage area. Putative hybrids occurred throughout the drainage at sites where both species were sampled.

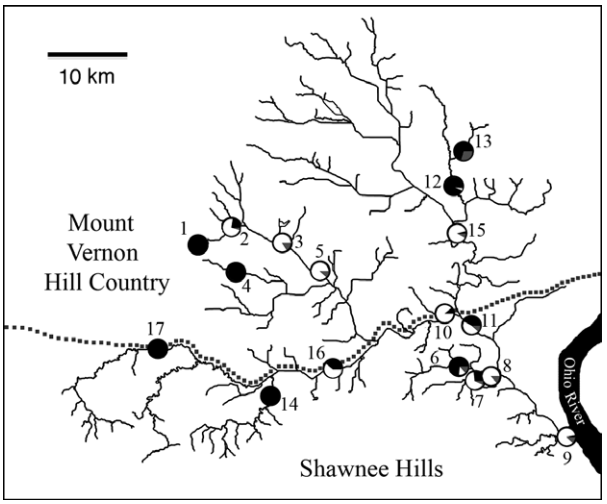

Fig. 1. Map of Saline River watershed showing study site locations with relative abundances. Black shows the proportion of *Fundulus olivaceus*, white shows the relative abundance of *F. notatus*, and gray indicates individuals of hybrid ancestry. The dashed line represents the boundary between Mount Vernon Hill Country to the north, and the Shawnee Hills region to the south.

Genetic structure

Only one locus in one population showed a significant deviation from Hardy–Weinberg equilibrium following a sequential Bonferroni correction (Rice 1989). Pairwise site F_{ST} values for both *F. notatus* and *F. olivaceus* are shown in Tables 3 and 4, respectively. A significant pattern of isolation by distance was found among *F. olivaceus* population samples ($r = 0.48$, $P = 0.005$), but not among *F. notatus* samples ($r = 0.12$, $P = 0.31$) (Fig. 2). This correlation

Table 2. Sample site identification numbers, drainage fork designation, latitude/longitude coordinates, numbers of individuals collected by species and cumulative drainage area.

Site ID	Fork	Latitude	Longitude	Number of *Fundulus olivaceus individuals*	Number of *F. notatus* individuals	Number of individuals of hybrid ancestry	Cumulative drainage area (km^2)
1	Middle	37.81	−88.76	27	0	0	3.8
2	Middle	37.86	−88.70	19	65	9	18.9
3	Middle	37.81	−88.61	0	19	5	181.1
4	Middle	37.79	−88.71	30	0	0	15.0
5	Middle	37.77	−88.54	0	6	0	215.3
6	Eagle	37.66	−88.32	14	2	2	65.8
7	Eagle	37.65	−88.30	6	11	2	58.7
8	Eagle	37.65	−88.26	6	14	3	148.6
9	Main	37.57	−88.13	1	29	3	3041.9
10	Main	37.73	−88.35	1	8	0	1437.0
11	Main	37.70	−88.29	10	12	4	2415.1
12	North	37.89	−88.33	42	2	1	132.9
13	North	37.94	−88.33	14	0	6	29.2
14	South	37.64	−88.63	41	0	0	84.7
15	North	37.83	−88.33	1	22	2	998.1
16	South	37.66	−88.63	28	38	0	657.5
17	South	37.68	−88.80	18	0	0	0.9

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Table 3. Genetic differentiation between *Fundulus notatus* populations (pairwise F_{ST}) among sites in the Middle fork, North fork and Eagle creek tributaries as well as the main stem of the Saline River.

Site	2	3	5	7	8	9	10	11	15
3	0.026								
3	0.066	0.039							
7	0.050	0.022	0.060						
8	0.040	0.010	0.061	0.012					
9	0.034	0.009	0.026	0.010	0.028				
10	0.060	0.035	0.072	0.059	0.027	0.028			
11	0.038	0.001	0.037	0.020	0.009	0.000	0.050		
15	0.100	0.149	0.176	0.141	0.119	0.130	0.165	0.144	
16	0.033	0.034	0.061	0.039	0.015	0.024	0.052	0.024	0.056

Table 4. Genetic differentiation between *Fundulus olivaceus* populations (pairwise F_{ST}) among sites in the Middle fork, North fork, South Fork and Eagle creek tributaries as well as the main stem of the Saline River.

Site	1	2	4	6	7	11	12	13	14	16
2	0.000									
4	0.088	0.071								
6	0.059	0.037	0.047							
7	0.112	0.082	0.057	0.000						
11	0.029	0.011	0.066	0.009	0.041					
12	0.071	0.053	0.038	0.014	0.016	0.022				
13	0.120	0.114	0.083	0.041	0.032	0.072	0.020			
14	0.081	0.062	0.049	0.032	0.031	0.025	0.025	0.062		
16	0.086	0.067	0.043	0.022	0.011	0.039	0.017	0.053	0.024	
17	0.078	0.056	0.046	0.014	0.028	0.005	0.014	0.037	0.015	0.037

was marginally improved in *F. olivaceus* when controlling for effects of habitat ($r^2 = 0.25$, $P = 0.002$). AMOVA revealed that 2.2% of variation within *F. olivaceus* was due to differences between forks ($P = 0.005$), 1.5% was due to differences between samples within forks ($P < 0.001$), and 96.4% was due to differences between individuals within sample sites ($P < 0.001$). AMOVA for *F. notatus* showed 1.3% of variation due to differences between forks ($P = 0.03$), 2.7% was due to differences between samples within forks ($P < 0.001$), and 96% was due to differences between individuals within sample sites ($P < 0.001$). Overall, F_{ST} values were small but significant in both species ($F_{ST} = 0.036$, $P < 0.001$ for *F. olivaceus*, and $F_{ST} = 0.040$, $P < 0.001$ for *F. notatus*).

Analysis of data by the STRUCTURE clustering algorithm suggested only one distinct genetic cluster within each species. Spatial autocorrelation analysis revealed a shift towards less relatedness among individuals than expected at random as distance increased between sites within both *F. notatus* and *F. olivaceus* populations (Fig. 3). *Fundulus notatus* individuals demonstrated significant positive relatedness in the 0–15 km distance class only, while *F. olivaceus* showed positive relatedness in both the 0–15 and 15–30 km classes. Beyond 30 km, both species showed significant negative relatedness values.

Habitat variables

The 23 sites used for microhabitat collection included the 17 sites sampled in this study, plus six additional sites for which genetic data were reported in the study by Duvernell and Schaefer (2013). Of these, 12 sites were dominated by *F. olivaceus* and 11 sites were dominated by *F. notatus*. The first three axes of the PCA explained 73% of between-site variation in microhabitat variables. As this investigation examined only two species, relative abundance of *F. notatus* was used for a measure of species composition. Cumulative drainage area showed a strong negative relationship with the first PC axis ($r = -0.57$) and a moderate positive relationship with the second PC axis ($r = 0.37$). Cumulative drainage area and interactions between cumulative drainage area and the first two PC axes were associated with the proportion of *F. notatus* (Table 5). The first PC axis was associated with relative abundance of *F. notatus* ($r^2 = 0.482$, $P < 0.001$) (Fig. 4a). The second and third PC axes did not significantly associate with relative abundance of species ($r^2 = 0.007$, $P = 0.49$ for PC2, $r^2 = 0.05$, $P = 0.07$ for PC3). Cumulative drainage area was associated with relative abundance of *F. notatus* ($r^2 = 0.221$, $P = 0.021$) (Fig. 4b). The first PC axis described relative stream position (high

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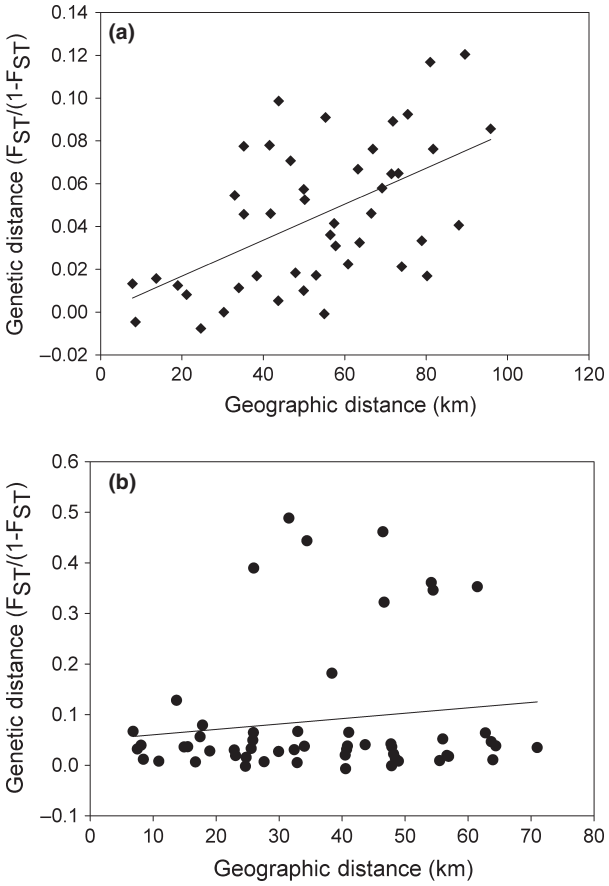

Fig. 2. Relationship between genetic distance and geographical distance. (a) *Fundulus olivaceus* (R^2 = 0.23, P = 0.005). (b) *Fundulus notatus* (R^2 = 0.01, P = 0.31).

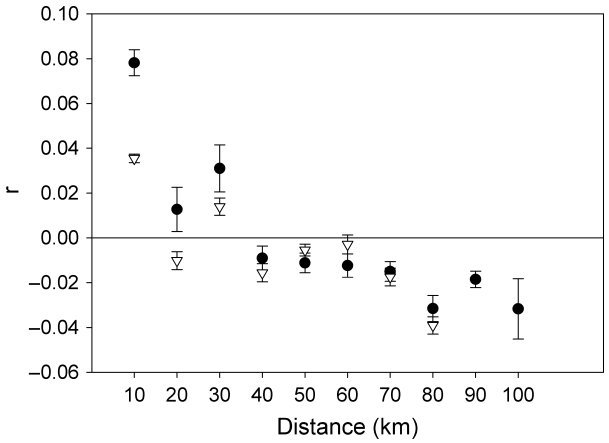

Fig. 3. Spatial autocorrelation analysis correlogram. Genetic correlation (*r*) of populations of *Fundulus olivaceus* (black) and *F. notatus* (white) decreases with increasing waterway distance classes.

loadings for woody in-stream cover and canopy cover) (Table 6). *F. notatus*-dominated sites had lower PC1 scores than *F. olivaceus*-dominated sites, although there was a region of overlap where sites dominated by either species showed similar PC1 scores (Fig. 5).

Table 5. Regression results for landscape and habitat variables and proportion of *Fundulus notatus*. Cumulative drainage area (CDA) positively associated with proportion of *F. notatus*. Interaction terms between CDA and habitat summary axes (PC1 and PC1) also were significantly associated with proportion of *F. notatus*.

	Partial correlation	*P*-value
Cumulative drainage area (CDA)	0.678	0.003
PC1	−0.360	0.156
PC2	0.376	0.137
CDA*PC1	−0.591	0.013
CDA*PC2	0.625	0.007
PC1*PC2	−0.344	0.177
CDA*PC1*PC2	−0.557	0.020

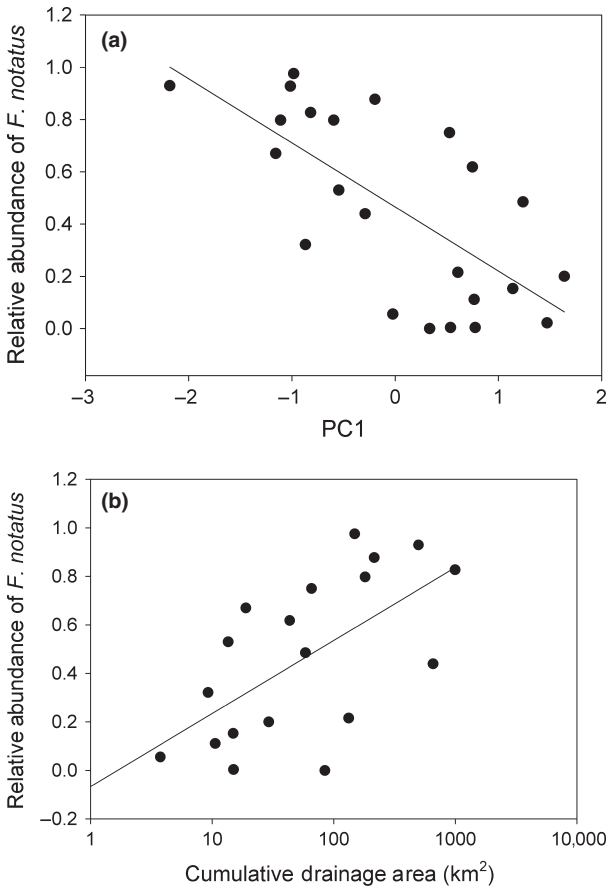

Fig. 4. Regression analyses of environmental data, species composition and cumulative drainage area. (a) Principal component axis 1 versus species composition expressed as proportion of *Fundulus notatus*. (b) Cumulative drainage area vs. species composition.

Discussion

We predicted that that the headwater specialist, *F. olivaceus,* would demonstrate a higher degree of population divergence and genetic structure relative to the larger river species, *F. notatus*, due to differences in within-drainage distributions. Evidence from isolation by distance and spatial autocorrelation

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Table 6. Principal component loadings table for the first three principal components. PC1 summarised bank characteristics and hydrological parameters (depth and flow), PC2 summarised substrate category, and PC3 summarised in-stream vegetation and canopy cover.

Parameter	PC1	PC2	PC3
Depth	0.798	−0.059	0.227
Flow	−0.665	0.403	−0.024
Substrate	0.131	−0.744	−0.556
Vegetation	0.195	−0.518	0.609
Woody cover	0.617	0.281	0.248
Bank stability	0.822	0.282	0.185
Bank slope	0.872	0.047	−0.095
Canopy cover	−0.176	−0.375	0.584

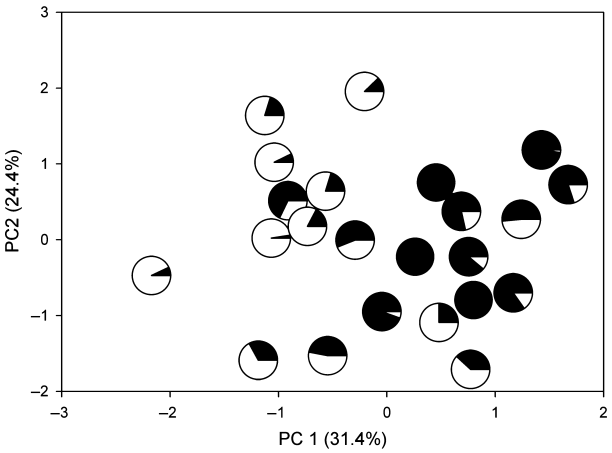

Fig. 5. Principal component plot showing the first two principal component axes. The admixture proportion of *Fundulus olivaceus* (black) and *F. notatus* (white) is indicated for each site.

analyses is consistent with *F. olivaceus* populations showing greater genetic isolation than *F. notatus* populations. Spatial autocorrelation analyses demonstrated significantly higher similarities between *F. olivaceus* compared with *F. notatus* individuals at distances up to 30 km. Likewise, population samples of both species exhibited similarly low estimates of F_{ST} and high relatedness scores at shorter distances (less than approximately 30 km, Fig. 2), but *F. olivaceus* population samples exhibited generally larger estimates of F_{ST} than *F. notatus* at longer distances. The spatial autocorrelation and isolation by distance patterns in *F. olivaceus* are consistent with high gene flow among sites within headwater habitats and reduced gene flow among populations separated among isolated headwater habitats.

Although values of F_{ST} were small in both species (0.036 for *F. olivaceus*, and 0.040 for *F. notatus*), they were statistically significant. Our observed F_{ST} values fell within ranges reported in taxa with stream distributions similar to *Fundulus* (darters: Haponski et al. 2009; shiners: Franssen 2012). At a regional scale, populations of the congener *F. sciadicus* showed high levels of F_{ST}, suggesting high levels of genetic structure among drainages (Li et al. 2009). Within the watershed, lack of isolation by distance pattern in *F. notatus* could be due to high levels of gene flow (Hutchison & Templeton 1999), although gene flow was not directly evaluated in this study. A subset of pairwise estimates among *F. notatus* population samples appeared as outliers with unusually high F_{ST} values (Fig. 2b). Each of these outliers included site 15, our only *F. notatus* sample from the North Fork of the Saline River (Fig. 1). An explanation for this apparent outlier *F. notatus* population sample will require additional sampling in the North Fork drainage. The *F. olivaceus* samples from the North Fork (sites 12 and 13, Fig. 1) did not exhibit the same outlier pattern.

Bayesian analysis of population structure using STRUCTURE (Pritchard et al. 2000) indicated no genetic structure in either species. The STRUCTURE model has been shown to perform poorly at levels of F_{ST} lower than 0.05 (Latch et al. 2006), which may explain why results from Bayesian analysis do not agree with data from Mantel tests or spatial autocorrelation.

The pattern of species distribution met expectations that *F. olivaceus* would dominate upstream sites with a transition to *F. notatus* in larger downstream habitats. In most drainages throughout the range of the two species, contact zones are typically disjunct and associated with tributary/river confluences (Duvernell & Schaefer 2013). In this study, *F. olivaceus* was found in nearly all sites throughout the drainage including the largest part of the Saline River immediately upstream of its confluence with the Ohio River. Likewise, *F. notatus* was found in a majority of sites, including all but the uppermost reaches of tributary streams where water flow was often intermittent. This distribution pattern agrees with previously published studies in the Saline River system and demonstrates that the contact zone between these two species is diffuse in all tributary forks in this drainage (Braasch & Smith 1965; Thomerson 1966).

Species distributions correlated with microhabitat parameters where *F. olivaceus*-dominated sites had shallow bank slopes, more in-stream cover and more canopy cover; and *F. notatus*-dominated sites had steep bank slopes, less in-stream cover and less canopy cover. Cumulative drainage area also correlated with species distribution and microhabitat factors. Our study agreed with results from Schaefer et al. (2011), which demonstrated that changes in cumulative drainage area between tributary and river levels are the best predictor of proportion of *Fundulus* species in contact zones. Cumulative drainage area differences upstream versus downstream of a confluence can act as a metric of amount of habitat change. Habitat parameters may be instructive in determination of

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suitable patches within regions of contact zones where both species occur in similar proportions. Differences in habitat preference may also drive patterns of genetic structure in these species.

The breadth of species distribution overlap in this drainage in all habitats could facilitate gene flow. In addition, *Fundulus* have the potential to exhibit large exploratory movements, which may result in homogenisation between populations. In a previous mark–recapture study, *F. notatus* and *F. olivaceus* individuals were shown to move hundreds of metres in a matter of a few days (Alldredge et al. 2011). Similar exploratory movements have been observed in both *F. notatus* and *F. olivaceus* in syntopy within the Saline River watershed (Duvernell, D.D., unpublished data). Therefore, high levels of gene flow could be the reason that we fail to detect genetic structure among these populations.

We saw evidence for very limited genetic structure in *Fundulus* populations within the Saline River watershed. We attribute interspecific differences in isolation by distance patterns to the distributions of the species within the dendritic stream system. Within this drainage, we also saw changes in patch characteristics that correlate with changes in relative abundance as we move from upstream to downstream. It is possible that patch quality drives the species distributions within this system and thus also drives the isolation by distance pattern of *F. olivaceus*. Species distributions and population genetic structure arise through interaction of multiple abiotic and biotic factors. Greater understanding of how these patterns arise can be gained by evaluation of the potential underlying drivers of diversification including movement patterns and patch preferences.

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